MODEL QUESTIONS

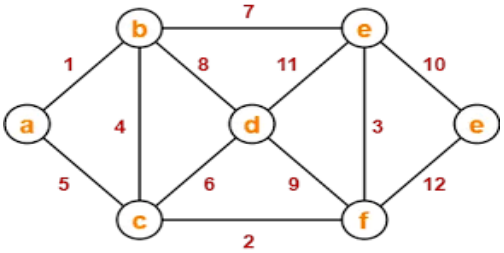
Short Questions(2 Mark)

1. Which one is better in terms of space complexity, Quick sort or Merge sort? Justify your answer.
2. Write the recurrence equation of merge sort.
3. What is the time complexity of a heapify( )?
4. Differentiate between Greedy & Dynamic Programming.
5. What is minimum spanning tree?
6. What is the time complexity of Prim's Algorithm?
7. What is the objective of Dijkstra Algorithm?
8. Which one is bigger n^2 or 2^n ? Justify.
9. Define Big O Notation.
10. Write the recurrence equation of quick sort.
11. *What is the time complexity of a binary search?*
12. *What will be the time complexity of merge sort if all the elements are already arranged in increasing order?*
13. *Define time complexity and space complexity of an algorithm.*
14. *What is the time complexity of Prim's Algorithm*?
15. What is Reducibility?
16. Ordering by asymptotic growth rates
    n^2, 2^n, n!, 1, n^3, n log n
17. Suppose the elements of an array are already sorted. Quick sort is applied on this array. Then Time complexity will be of best case or worst case. Justify your answer.
18. Write two applications of Huffman coding.
19. Differentiate between P & NP Problems.
20. What is the time complexity of Krushkal Algorithm?
21. What is the disadvantages of Dijkstra Algorithm?
22. What is the time complexity of bellman ford algorithm?
23. Differentiate between max heapify and min heapify?
24. Define divide and conquer approach. State two examples.
25. If the graph contains 6 nodes then how many maximum spanning tree is possible?
26. State different states of algorithm.

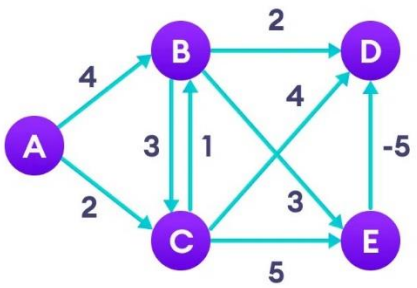
27. What are the various properties of an algorithm?
28. Define recurrence relation. What are the methods to solve recurrence relation?
29. What is single source shortest path algorithm?

Long Question(8 Mark)

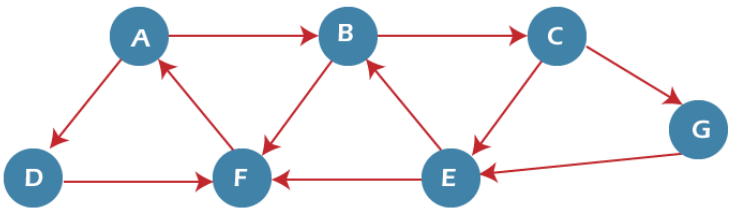
1. Write the linear search Algorithm and Analyze its Best, Worst and Average case time complexity.
2. Explain MERGE SORT for following elements: 27, 18, 19, 14, 16, 23, 21, 12
3. Write an algorithm for QUICK SORT.
4. Solve the knapsack problem for n = 4 , W=40, (p1,p2,p3,p4) = (11,21,31,33) (w1,w2,w3,w4) = (2,11,22,15)
5. Find Huffman code for a - 4 , b - 5 , c - 7, d - 8 , e - 10, f - 12 , g – 20.
6. Explain how Kruskal's algorithm can be used to find minimum spanning tree of a graph with an example.
7. Explain Prim's algorithm with an example.
8. Write an algorithm for Activity selection problem. Solve the problem where start times = (1,3,0,5,3,5,6,8,8,2,12) and Finish times = (4,5,6,7,8,9,10,11,12,13,14).
9. Create max-heap for following elements: 26, 18, 30, 32, 16, 74. Draw diagrams by inserting one element at a time.
10. Write the linear search Algorithm and analyze its Best, Worst and Average case time complexity.
11. Explain QUICK SORT for following elements: 8, 7, 11, 6, 4, 5, 9.
12. Write an algorithm for Knapsack problem.
13. Explain Prim's algorithm with an example.
14. Write Dijkstra Algorithm.
15. Create max-heap for following elements: 3, 2, 4 , 5, 1, 6 , 7. Draw diagrams by inserting one element at a time.
16. Write an algorithm for Binary search and analyze its Best, Worst and Average case time complexity.
17. Explain how Kruskal's algorithm can be used to find minimum spanning tree of a graph given below:



18. Write Krushkal's algorithm to find minimum spanning tree.
19. Find LCS for X = ABABDC and Y = BCBCABAD. Write an algorithm for LCS.
20. *Find an optimal parenthesization of a matrix chain multiplication whose sequence of dimensions (orders) are < 10, 20, 50, 1, 100 >*
21. Find out shortest path for the following graph using Bellman ford algorithm



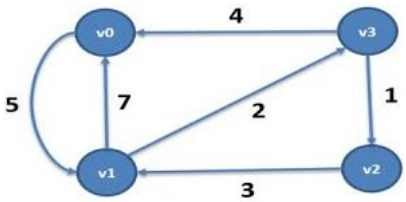
22. *Write algorithms for BFS and DFS. Find BFS and DFS where adjacent list is given below. The starting vertex is 'A'*



Adjacency Lists

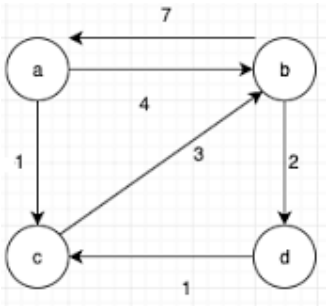
A : B, D
B : C, F
C : E, G
G : E
E : B, F
F : A
D : F

23. *Describe Floyd Warshall's algorithm for all pairs shortest path problem. Apply the same to the following Graph and discuss its time complexity.*



24. *Find an optimal parenthesization of a matrix chain multiplication whose sequence of dimensions (orders)are < 4, 10, 3, 12, 20, 7 >*

25. *Describe Floyd Warshall algorithm for all pairs shortest path problem. Apply the same to the following Graph and discuss its time complexity.*



26. You are given 4 matrices: A1: 10×5, A2: 5×20, A3: 20×10, A4: 10×5

    We need to find the optimal parenthesization for computing A1 A2 A3 A4 that minimizes the number of operations.